Class Exercise: Doppler Effect

Consider a train moving at 40 m/s and it has a 100 Hz horn. During the motion of this train, the air is mostly still and the speed of sound is about 340 m/s.

i) What sound frequencies are detected by a person standing at the side of the tracks a) as the train approaches and b) after it passes.

ii) What sound frequency is observed by the train's engineer travelling on the train.

Solution:

Strategy → use $f' = f\left[\frac{v}{v \pm v_s}\right]$

for moving source.

- (-) sign ⇒ approaching train
- (+) sign ⇒ receding train.

- There are 2 Doppler Shifts:
  1. moving source
  2. moving detector

Solution :

i) • Approaching Train

$$f' = f\left[\frac{v}{v - v_s}\right]$$

$$= (100\ Hz)\left[\frac{340\ m/s}{(340 - 40)\ m/s}\right]$$

$$= 113.33\ Hz\ \#$$

• Receding Train

$$f' = f\left[\frac{v}{v + v_s}\right]$$

$$= (100\ Hz)\left[\frac{340\ m/s}{(340 + 40)\ m/s}\right]$$

$$= 89.47\ Hz\ \#$$

ii)

- The engineer should observe the same frequency as emitted by the horn on the train.

$$f = 100\ Hz.$$